



Scenario

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	Our organization's internal network were not able to access any network resources. This happened because of DoS attack where there was incoming flood of ICMP packets from the malicious actor. The incident managemet team responded by blocing incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services. This attack compromised the internal network for two hours until it was resolved.
Identify	The cybersecurity team investigated this security event and found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.
Protect	To address this the network security team implemented a new firewall rule to limit the rate of incoming ICMP packets, to check for spoofed IP addresses on incoming ICMP packets by source IP address verification on the firewall, to detect abnormal traffic patterns using network monitoring software, and to filter out some ICMP traffic based on suspicious characteristics using an IDS/IPS system.

Detect	To detect new attacks in the future, the team will use a network monitoring system and an intrusion detection system to monitor all the incoming traffic from the internet.
Respond	The cybersecurity team can contain the incident by configuring the firewall like blocking the incoming ICMP packets and take measures like stopping non-critical network services offline so that business continues. Using network monitoring system and IDS/IPS we can identify security risks earliest as possible.
Recover	To recover early as possible the cybersecurity team can implement framework like Risk Management Framework so that when any attack occurs we can be prepared by assigning roles, following guidelines, selecting tools like firewall, IDS/IPS and implementing them.

Reflections/Notes:
